

Top Quark Signal in the Cosmological Higgs Collider

1 Two-Weyl setup and Feynman rules

1.1 Lagrangian, basis and mode functions

We start with the top sector of the Standard Model before choosing a coordinate system. In the usual left/right notation, let $Q_{3L} = (t_L, b_L)^T$ and $\tilde{\Phi} = (\Phi^{2*}, -\Phi^{1*})^T$. Its relevant curved-space action is, in four-component chiral notation,

$$S_{\text{top}} = \int d^4x \sqrt{-g} \left[i\bar{Q}_{3L}\gamma^a e_a{}^\mu D_\mu Q_{3L} + i\bar{t}_R\gamma^a e_a{}^\mu D_\mu t_R - y_t (\bar{Q}_{3L}\tilde{\Phi} t_R + \bar{t}_R\tilde{\Phi}^\dagger Q_{3L}) \right].$$

To express everything with left-handed two-component spinors, identify each chiral field with its nonzero Weyl component and replace the right-handed t_R by its charge conjugate:

$$Q_3 = (t_{\text{cov}}, b_{\text{cov}})^T, \quad t_{\text{cov}} = t_L, \quad (t_R)^C = C\bar{t}_R^T, \quad t_{\text{cov}}^c = [(t_R)^C]_L.$$

Here C is the charge-conjugation matrix. Equivalently, for the two-component right-handed spinor, $t_{\text{cov}}^c = i\sigma^2 t_R^*$, with σ^2 acting on spinor indices. Thus t^c is the same right-handed top degree of freedom written as a left-handed charge-conjugate field. Its SM gauge charges are conjugated, so its covariant derivative acts in the conjugate representation. In particular,

$$\begin{aligned} \bar{t}_R\tilde{\Phi}^\dagger Q_{3L} &\longleftrightarrow (\Phi^2 t_{\text{cov}} - \Phi^1 b_{\text{cov}}) t_{\text{cov}}^c = \epsilon_{ij} Q_3^i \Phi^j t_{\text{cov}}^c, \\ i\bar{t}_R\gamma^a e_a{}^\mu D_\mu t_R &\longleftrightarrow i t_{\text{cov}}^{c\dagger} \bar{\sigma}^a e_a{}^\mu D_\mu t_{\text{cov}}^c, \end{aligned}$$

where $\epsilon_{12} = +1$ and the kinetic identity holds up to a total derivative. We use the phase of t_{cov}^c for which the Yukawa term has the sign below. The action therefore becomes

$$S_{\text{top}} = \int d^4x \sqrt{-g} \left[iQ_3^\dagger \bar{\sigma}^a e_a{}^\mu D_\mu Q_3 + i t_{\text{cov}}^{c\dagger} \bar{\sigma}^a e_a{}^\mu D_\mu t_{\text{cov}}^c - y_t (\epsilon_{ij} Q_3^i \Phi^j t_{\text{cov}}^c + \text{h.c.}) \right]. \quad (1)$$

Here Φ is the SM Higgs doublet, D_μ contains the spin and SM gauge connections, and $\sigma^a = (\mathbf{1}_2, \boldsymbol{\sigma})$, $\bar{\sigma}^a = (\mathbf{1}_2, -\boldsymbol{\sigma})$ with $\eta_{ab} = \text{diag}(1, -1, -1, -1)$. Gauge backgrounds are set to zero in the Yukawa box considered below. Choosing the neutral Higgs background $\Phi = (0, (h_0 + \delta h)/\sqrt{2})^T$ selects the top component of Q_3 . We assume h_0 is constant, and write

$$g_t = \frac{y_t}{\sqrt{2}}, \quad m_t = g_t h_0, \quad \mu = \frac{m_t}{H}, \quad N_c = 3. \quad (2)$$

The two-component contraction $t_{\text{cov}} t_{\text{cov}}^c$ includes its antisymmetric spinor epsilon tensor. Since the fields are distinct, the Yukawa term has no factor of $1/2$.

In conformal de Sitter coordinates,

$$ds^2 = a(\tau)^2(d\tau^2 - d\mathbf{x}^2), \quad a(\tau) = -\frac{1}{H\tau}, \quad \tau < 0, \quad e_a{}^\mu = a^{-1}\delta_a{}^\mu. \quad (3)$$

Rescale the two Weyl fields by their conformal weight,

$$t = a^{3/2}t_{\text{cov}}, \quad t^c = a^{3/2}t_{\text{cov}}^c. \quad (4)$$

The tetrad and spin connection give $ia^3 t_{\text{cov}}^\dagger \bar{\sigma}^\mu (\partial_\mu + 3a'\delta_\mu^0/2a)t_{\text{cov}}$, and likewise for t_{cov}^c . The extra $3a'/2a$ cancels the derivative of $a^{-3/2}$ in Eq. (4). The Yukawa bilinear instead retains one factor of $a^4(a^{-3/2})^2 = a$. Thus the action for the top sector becomes

$$S = \int d\tau d^3x \left\{ it^\dagger \bar{\sigma}^\mu \partial_\mu t + it^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(\tau)(m_t + g_t \delta h)(tt^c + t^\dagger t^{c\dagger}) \right\}. \quad (5)$$

All fermion fields from here onward are the rescaled fields. The background mass m_t belongs to the free equation; each δh insertion supplies one Yukawa vertex.

To keep the two Weyl equations in one matrix equation, introduce the four-component Dirac column

$$\Psi_W = \begin{pmatrix} t_\alpha \\ t^{c\dagger\dot{\alpha}} \end{pmatrix}, \quad \beta_W = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}. \quad (6)$$

The epsilon contractions in Eq. (5) are understood in this column notation, so its mass bilinear is $-am_t \Psi_W^\dagger \beta_W \Psi_W$ in our index convention. Under top number, $t \mapsto e^{i\vartheta} t$ and $t^c \mapsto e^{-i\vartheta} t^c$. Both entries of Ψ_W therefore have the same charge; no Majorana reality condition is imposed.

The mass matrix is diagonal in a simple orthonormal component basis. Rather than naming its two basis columns separately, define

$$U_B = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{1}_2 & -\mathbf{1}_2 \\ \mathbf{1}_2 & \mathbf{1}_2 \end{pmatrix}, \quad \Psi = U_B^\dagger \Psi_W, \quad \beta = U_B^\dagger \beta_W U_B = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}. \quad (7)$$

This unitary change acts only on spinor components. It preserves the top charge, canonical anticommutators, and traces.

Use the Fourier convention

$$\Psi_W(\tau, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{x}} \Psi_W(\tau, \mathbf{k}), \quad \partial_i \longmapsto ik_i. \quad (8)$$

Varying the quadratic action and then applying U_B gives, with the same momentum on both sides,

$$\begin{aligned} i\partial_\tau \Psi_W(\tau, \mathbf{k}) &= \begin{pmatrix} -\boldsymbol{\sigma} \cdot \mathbf{k} & am_t \mathbf{1}_2 \\ am_t \mathbf{1}_2 & \boldsymbol{\sigma} \cdot \mathbf{k} \end{pmatrix} \Psi_W(\tau, \mathbf{k}), \\ i\partial_\tau \Psi(\tau, \mathbf{k}) &= \begin{pmatrix} am_t \mathbf{1}_2 & \boldsymbol{\sigma} \cdot \mathbf{k} \\ \boldsymbol{\sigma} \cdot \mathbf{k} & -am_t \mathbf{1}_2 \end{pmatrix} \Psi(\tau, \mathbf{k}). \end{aligned} \quad (9)$$

The matrix in the second line is Hermitian; at fixed time its square is $(k^2 + a^2 m_t^2) \mathbf{1}_4$.

For nonzero momentum, write $k = |\mathbf{k}| > 0$ and $\mathbf{k} = k(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$. We use the following explicit normalized helicity basis:

$$\xi_+(\mathbf{k}) = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \quad \xi_-(\mathbf{k}) = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}. \quad (10)$$

They obey

$$(\boldsymbol{\sigma} \cdot \hat{\mathbf{k}}) \xi_h = h \xi_h, \quad \xi_h^\dagger \xi_{h'} = \delta_{hh'}, \quad \sum_{h=\pm 1} \xi_h \xi_h^\dagger = \mathbf{1}_2. \quad (11)$$

The phases in Eq. (10) are a convention; the subsequent propagators depend only on the helicity projectors. For fixed nonzero momentum, “positive frequency” means a mode approaching $e^{-ik\tau}$ in the far past $\tau \rightarrow -\infty$; “negative frequency” means one approaching $e^{+ik\tau}$. Indeed, $i\partial_\tau e^{\mp ik\tau} = \pm k e^{\mp ik\tau}$. The superscript (+) or (−) on a mode denotes this frequency, while $h = \pm 1$ denotes helicity. Neither sign is an SK branch label. We first solve for the positive-frequency mode:

$$\Psi_h^{(+)}(\tau, \mathbf{k}) = \begin{pmatrix} F(x) \xi_h \\ h G(x) \xi_h \end{pmatrix}, \quad x = -k\tau > 0. \quad (12)$$

The factor h in the lower entry makes the same two radial functions work for both helicities. Since $am_t = \mu k/x$ and $\partial_\tau = -k\partial_x$, Eq. (9) reduces to

$$-iF' = \frac{\mu}{x} F + G, \quad -iG' = F - \frac{\mu}{x} G, \quad (13)$$

$$F'' + \left[1 + \frac{\mu^2 + i\mu}{x^2} \right] F = 0, \quad G'' + \left[1 + \frac{\mu^2 - i\mu}{x^2} \right] G = 0. \quad (14)$$

The second line follows by eliminating one component from the first. These are Bessel equations with orders $1/2 - i\mu$ and $1/2 + i\mu$. The positive-frequency BD condition is $e^{-ik\tau} = e^{ix}$, so we choose Hankel functions of the first kind. Normalizing the positive-frequency column to unit norm fixes

$$\boxed{F(x) = \frac{\sqrt{\pi x}}{2} e^{\pi\mu/2} H_{1/2-i\mu}^{(1)}(x), \quad G(x) = \frac{\sqrt{\pi x}}{2} e^{-\pi\mu/2} H_{1/2+i\mu}^{(1)}(x).} \quad (15)$$

The relative coefficients in this pair follow directly from $G = -iF' - \mu F/x$ and the Hankel identities $H_{\nu-1}^{(1)} = H_\nu^{(1)'} + \nu H_\nu^{(1)}/x$ and $H_{-1/2-i\mu}^{(1)} = i e^{-\pi\mu} H_{1/2+i\mu}^{(1)}$. Both functions approach $-ie^{ix}/\sqrt{2}$ as $x \rightarrow \infty$. The first-order equations imply $\partial_\tau(|F|^2 + |G|^2) = 0$, so this asymptotic normalization gives $|F|^2 + |G|^2 = 1$ at every time. The common phase is conventional; the relative phase between F and G is fixed by Eq. (13). The independent negative-frequency mode at the same Fourier momentum is then

$$\Psi_h^{(-)}(\tau, \mathbf{k}) = \begin{pmatrix} G(x)^* \xi_h \\ -h F(x)^* \xi_h \end{pmatrix} \propto e^{+ik\tau} \quad (\tau \rightarrow -\infty). \quad (16)$$

Here $*$ means complex conjugation of the time-dependent functions; the relative minus sign makes this column solve the same field equation at momentum $\mathbf{k}$.

1.2 SK propagators

Here $A, B = 1, \dots, 4$ label components of the four-component spinor Ψ ; they are not SK indices. The SK branch labels will be $s, s' \in \{+, -\}$. All propagators in this subsection are first computed in the rotated Ψ basis. Outer products of the modes give the two Wightman functions, and contour ordering then gives the four SK propagators. The positive- and negative-frequency solutions can be collected into two 4×2 matrices. Their two-dimensional column space represents the two spin states; multiplying either matrix by ξ_h selects the helicity- h solution:

$$u(\mathbf{k}, \tau) = \begin{pmatrix} F(x)\mathbf{1}_2 \\ G(x)\boldsymbol{\sigma} \cdot \hat{\mathbf{k}} \end{pmatrix}, \quad v(\mathbf{k}, \tau) = \begin{pmatrix} G(x)^*\mathbf{1}_2 \\ -F(x)^*\boldsymbol{\sigma} \cdot \hat{\mathbf{k}} \end{pmatrix}. \quad (17)$$

Thus $u\xi_h = \Psi_h^{(+)}$ and $v\xi_h = \Psi_h^{(-)}$. At fixed Fourier momentum the field operator expands as

$$\begin{aligned} \Psi(\tau, \mathbf{k}) = & \sum_{h=\pm 1} \begin{pmatrix} F(x)\xi_h(\mathbf{k}) \\ hG(x)\xi_h(\mathbf{k}) \end{pmatrix} b_h(\mathbf{k}) \\ & + \sum_{h=\pm 1} \begin{pmatrix} G(x)^*\xi_h(\mathbf{k}) \\ -hF(x)^*\xi_h(\mathbf{k}) \end{pmatrix} d_h^\dagger(-\mathbf{k}). \end{aligned} \quad (18)$$

For each h , $b_h(\mathbf{k})$ annihilates a particle and $d_h^\dagger(-\mathbf{k})$ creates an antiparticle. The label h on the creation operator refers to the basis spinor $\xi_h(\mathbf{k})$ used in this Fourier expansion. The $-\mathbf{k}$ argument follows from Eq. (8); color labels are suppressed. The compact form of Eq. (18) is $\Psi = ub + vd^\dagger$, where $b(\mathbf{k}) = \sum_h \xi_h(\mathbf{k})b_h(\mathbf{k})$ and $d^\dagger(-\mathbf{k}) = \sum_h \xi_h(\mathbf{k})d_h^\dagger(-\mathbf{k})$ are two-entry operator columns. The dagger on an operator denotes its adjoint, whereas $*$ on F, G means complex conjugation and $u^\dagger, v^\dagger$ mean Hermitian conjugation of matrices. The canonical anticommutators, together with the mode normalization, give $u^\dagger u = v^\dagger v = \mathbf{1}_2$, $u^\dagger v = 0$, and $uu^\dagger + vv^\dagger = \mathbf{1}_4$. The modes in the original Weyl basis are $U_B u$ and $U_B v$.

Before choosing an SK branch, $R^>$ and $R^<$ denote the two possible field orderings; the symbols $>$ and $<$ do not compare τ with τ' . Matrix multiplication sums over the two spin states. We define these contractions without an extra prefactor of i , and suppress the momentum-conserving delta function. With the first argument at the field end of the ordered line,

$$\begin{aligned} R_{AB}^>(\mathbf{k}; \tau, \tau') &= \langle \Psi_A(\tau, \mathbf{k}) \Psi_B^\dagger(\tau', \mathbf{k}) \rangle = [u(\tau)u^\dagger(\tau')]_{AB}, \\ R_{AB}^<(\mathbf{k}; \tau, \tau') &= -\langle \Psi_B^\dagger(\tau', \mathbf{k}) \Psi_A(\tau, \mathbf{k}) \rangle = -[v(\tau)v^\dagger(\tau')]_{AB}. \end{aligned} \quad (19)$$

Here is the vacuum contraction that gives the second equality in each line. Let $|0\rangle$ be the BD vacuum, so $b_h(\mathbf{k})|0\rangle = d_h(\mathbf{k})|0\rangle = 0$. Taking the adjoint of Eq. (18) gives

$$\Psi_B^\dagger(\tau', \mathbf{k}') = \sum_{h'=\pm 1} \left[b_{h'}^\dagger(\mathbf{k}') \Psi_{h',B}^{(+)*}(\tau', \mathbf{k}') + d_{h'}^\dagger(-\mathbf{k}') \Psi_{h',B}^{(-)*}(\tau', \mathbf{k}') \right]. \quad (20)$$

The only nonzero operator products between these vacuum states are

$$\langle 0|b_h(\mathbf{k})b_{h'}^\dagger(\mathbf{k}')|0\rangle = \langle 0|d_h(-\mathbf{k})d_{h'}^\dagger(-\mathbf{k}')|0\rangle = (2\pi)^3 \delta_{hh'} \delta^3(\mathbf{k} - \mathbf{k}'). \quad (21)$$

Substituting Eqs. (18) and (20) into the two expectation values leaves one operator product in each ordering. Suppressing the arguments of the modes and operators for this intermediate step,

$$\langle 0|\Psi_A \Psi_B^\dagger|0\rangle = \sum_{h,h'} \Psi_{h,A}^{(+)} \Psi_{h',B}^{(+)*} \langle 0|b_h b_{h'}^\dagger|0\rangle,$$

$$-\langle 0 | \Psi_B^\dagger \Psi_A | 0 \rangle = - \sum_{h,h'} \Psi_{h',B}^{(-)*} \Psi_{h,A}^{(-)} \langle 0 | d_{h'} d_h^\dagger | 0 \rangle. \quad (22)$$

Terms with $b^\dagger b$, $d^\dagger d$, or a mixed b, d product vanish in the vacuum. Restoring the arguments and using Eq. (21) gives

$$\begin{aligned} \langle 0 | \Psi_A(\tau, \mathbf{k}) \Psi_B^\dagger(\tau', \mathbf{k}') | 0 \rangle &= (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}') \\ &\quad \times \sum_h \Psi_{h,A}^{(+)}(\tau, \mathbf{k}) \Psi_{h,B}^{(+)*}(\tau', \mathbf{k}'), \\ -\langle 0 | \Psi_B^\dagger(\tau', \mathbf{k}') \Psi_A(\tau, \mathbf{k}) | 0 \rangle &= -(2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}') \\ &\quad \times \sum_h \Psi_{h,A}^{(-)}(\tau, \mathbf{k}) \Psi_{h,B}^{(-)*}(\tau', \mathbf{k}'). \end{aligned} \quad (23)$$

Finally, Eq. (11) converts the helicity sums into matrix products. The common momentum $\mathbf{k}$ is suppressed in this display:

$$\begin{aligned} \sum_h \Psi_h^{(+)}(\tau) \Psi_h^{(+)\dagger}(\tau') &= u(\tau) \left(\sum_h \xi_h \xi_h^\dagger \right) u^\dagger(\tau') = u(\tau) u^\dagger(\tau'), \\ \sum_h \Psi_h^{(-)}(\tau) \Psi_h^{(-)\dagger}(\tau') &= v(\tau) \left(\sum_h \xi_h \xi_h^\dagger \right) v^\dagger(\tau') = v(\tau) v^\dagger(\tau'). \end{aligned} \quad (24)$$

These are precisely the products in Eq. (19) after removing the common momentum delta function. Both nonzero vacuum products in Eq. (21) are positive; the minus sign in $R^<$ comes from the fermionic ordering convention in its definition, not from the $dd^\dagger$ vacuum product.

Write $F_1 = F(-k\tau)$, $F_2 = F(-k\tau')$, and likewise for G ; also write $\sigma_k = \boldsymbol{\sigma} \cdot \hat{\mathbf{k}}$. Multiplication of the matrices in Eq. (17) then gives

$$R^> = \begin{pmatrix} F_1 F_2^* \mathbf{1}_2 & F_1 G_2^* \sigma_k \\ G_1 F_2^* \sigma_k & G_1 G_2^* \mathbf{1}_2 \end{pmatrix}, \quad R^< = \begin{pmatrix} -G_1^* G_2 \mathbf{1}_2 & G_1^* F_2 \sigma_k \\ F_1^* G_2 \sigma_k & -F_1^* F_2 \mathbf{1}_2 \end{pmatrix}. \quad (25)$$

For example, the upper-right block of $R^<$ is positive because $-(G_1^* \mathbf{1}_2)(-F_2^* \sigma_k)^\dagger = G_1^* F_2 \sigma_k$. This is a useful check on the relative sign of the two components of v .

Let $+$ denote the forward SK branch and $-$ the backward branch. In $R_{ss'}(\mathbf{k}; \tau, \tau')$, s belongs to the first time τ and s' to the second time τ' . Each $R_{ss'}$ remains a 4×4 spinor matrix with entries $R_{ss';AB}$, and θ is the step function. Contour ordering gives

$$\begin{aligned} R_{-+} &= R^>, & R_{+-} &= R^<, \\ R_{++} &= \theta(\tau - \tau') R^> + \theta(\tau' - \tau) R^<, & R_{--} &= \theta(\tau' - \tau) R^> + \theta(\tau - \tau') R^<. \end{aligned} \quad (26)$$

Thus R_{++} is time ordered and R_{--} is reverse time ordered. A line in the original Weyl basis uses $R_W = U_B R U_B^\dagger$. Each displayed entry of Eq. (25) is a 2×2 block, and its diagonal blocks are not $\langle tt \rangle$ or $\langle t^c t^c \rangle$. In the original Weyl basis, the diagonal blocks of R_W contain $\langle tt^\dagger \rangle$ and $\langle t^{c\dagger} t^c \rangle$; the off-diagonal blocks contain $\langle tt^c \rangle$ and its conjugate, with the ordering signs specified by $R^>$ or $R^<$. The Fourier convention in Eq. (8) fixes the momentum of a reversed line to $-\mathbf{k}$.

1.3 Feynman rules

The external Higgs fluctuation is treated as massless. Its bulk-to-boundary contraction on SK branch $s = \pm 1$ is

$$G_s(k; \tau) = \frac{H^2}{2k^3} (1 - isk\tau) e^{isk\tau}. \quad (27)$$

All external momenta are incoming, $\sum_i \mathbf{k}_i = 0$, and a prime on a correlator removes the overall $(2\pi)^3 \delta^3(\sum_i \mathbf{k}_i)$. We work in exact de Sitter with constant m_t and zero chemical potential, and keep the leading collapsed expansion. This is a spectator-background approximation: over the times sampled by the correlator we require $|\dot{h}_0| \ll H|h_0|$ and $|m_{h,\text{eff}}^2| \ll H^2$, with the renormalized Higgs effective mass including potential and curvature contributions. These conditions do not follow from a nonzero h_0 alone. The calculation gives the connected δh four-point function at order g_t^4 ; converting it to a curvature trispectrum requires a specified Higgs-to-curvature transfer mechanism. Since the Yukawa coupling is linear in δh , there is no elementary $\delta h^2 \bar{t} t$ vertex and hence no Yukawa bubble or triangle at this order with four external Higgs legs.

The following rules refer to the conformally rescaled fermions in Eq. (5). SK contractions contain no extra factor of i . The branch sign at a vertex will be denoted by $s = \pm 1$ here to distinguish it from the scale factor $a(\tau)$.

1. **External Higgs line.** Attach $G_s(k; \tau)$ from Eq. (27) to a vertex on branch s . External Higgs fields are labelled before summing permutations.
2. **Yukawa vertex.** From $S_{\text{int}} = -g_t \int d\tau d^3x a(\tau) \delta h \Psi_W^\dagger \beta_W \Psi_W$, the vertex matrix is

$$-is g_t a(\tau) (\beta_W)_{AB} \delta_{cc'} = -is g_t a(\tau) \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}_{AB} \delta_{cc'}. \quad (28)$$

Here A labels the $\Psi_W^\dagger$ slot, B the Ψ_W slot, and c, c' are color indices. The two off-diagonal blocks are precisely the $\delta h t t^c$ and $\delta h t^\dagger t^{\dagger c}$ vertices with the index positions specified below. Each has coefficient $-is g_t a(\tau)$; there is no $1/2$ since the two Weyl fields are distinct. In the mode basis, replace β_W by $\beta = \text{diag}(\mathbf{1}_2, -\mathbf{1}_2)$. There is only one power of the scale factor at a vertex. The background mass is already included in the propagator, so it is not inserted again.

3. **Fermion line.** A line with field end (A, τ, s) and dagger end (B, τ', s') contributes

$$[R_{W,ss'}(\mathbf{k}; \tau, \tau')]_{AB} \delta_{cc'}. \quad (29)$$

where $R_W = U_B R U_B^\dagger$ and R is given in Eqs. (25)–(26). The block dictionary is

$$R_{W,ss'} = \begin{pmatrix} \langle t_\alpha t_\beta^\dagger \rangle_{ss'} & \langle t_\alpha t^{c\beta} \rangle_{ss'} \\ \langle t^{c\dagger\alpha} t_\beta^\dagger \rangle_{ss'} & \langle t^{c\dagger\alpha} t^{c\beta} \rangle_{ss'} \end{pmatrix}. \quad (30)$$

Thus the epsilon tensors needed to raise the t^c indices are already included; one must not add another epsilon tensor to the vertex matrix. This fixes the normal and anomalous Weyl contractions in one convention. The charge arrow runs from the dagger end to the field end,

and the Fourier factor of this ordered line is $e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}$. If its endpoints are exchanged, use the coordinate-space Grassmann identity

$$\langle T_C \Psi_B^\dagger(\tau, \mathbf{x}) \Psi_A(\tau', \mathbf{x}') \rangle_{ss'} = -[R_{s's}(\tau', \mathbf{x}'; \tau, \mathbf{x})]_{AB}. \quad (31)$$

The coefficient of $e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}$ on the right is $-[R_{s's}(-\mathbf{k}; \tau', \tau)]_{AB}$. This distinguishes the Fourier transform of the adjoint field from the adjoint of a Fourier coefficient, whose momentum convention was used in Eq. (19). In particular the minus sign of $R^<$ in Eq. (19) is part of the line rule, not an optional extra sign. Charge-forbidden contractions such as $\langle tt \rangle$ and $\langle t^c t^c \rangle$ vanish.

4. **Integrals and closed loop.** Integrate every vertex over $d\tau \in (-\infty, 0)$, impose spatial momentum conservation at each vertex, and integrate an independent loop momentum with $d^3q/(2\pi)^3$. Use the BD $i\epsilon$ prescription for the early-time endpoint, and sum all vertex branch labels. A closed fermion loop gives -1 , a spin trace, and a color trace $N_c = 3$. No spin average or additional factor of two is included. Each labelled directed box below has symmetry factor one.

Multiplying these rules gives Eq. (43); its overall sign is $(-1)_{\text{loop}}(-i)^4 = -1$, with the product of the four branch signs kept inside the SK sum.

The calculation now follows the same order as the gauge-boson box note: identify the two soft lines, keep the other two lines as exact SK hard subgraphs, perform the spin or polarization contraction, and reduce the remaining loop integral to scalar F_0 convolutions. For fermions the polarization tensors are replaced by the matrices βR and a Pauli trace. The symmetric exchange of the external Higgs legs removes the leading equal-branch spin contraction, so the hard subgraphs must be expanded one order further before the scalar convolution is taken.

1.4 Nonlocal part of the soft propagator

For $\tau < 0$, the late-time argument is $x = -k\tau > 0$. Define only one coefficient,

$$A = -\frac{i2^{-1/2-i\mu}}{\sqrt{\pi}}\Gamma(\tfrac{1}{2} - i\mu), \quad c = -A^2 = \frac{2^{-1-2i\mu}}{\pi}\Gamma(\tfrac{1}{2} - i\mu)^2, \quad |c| = \frac{1}{2\cosh \pi\mu}. \quad (32)$$

After the factors $e^{\pm\pi\mu/2}$ are displayed explicitly, the two leading mode coefficients are A and $-A^*$; no second symbol is needed. The late-time modes are

$$F(-k\tau) = e^{\pi\mu/2}A(-k\tau)^{i\mu} - \frac{ie^{-\pi\mu/2}A^*}{1-2i\mu}(-k\tau)^{1-i\mu} + O((k\tau)^2 \text{ times the branches}), \quad (33)$$

$$G(-k\tau) = -e^{-\pi\mu/2}A^*(-k\tau)^{-i\mu} + \frac{ie^{\pi\mu/2}A}{1+2i\mu}(-k\tau)^{1+i\mu} + O((k\tau)^2 \text{ times the branches}). \quad (34)$$

In particular,

$$R_{ab}^{\text{NL}}(\mathbf{k}; \tau, \tau') = c(k^2\tau\tau')^{i\mu} \begin{pmatrix} 0 & \sigma_k \\ 0 & 0 \end{pmatrix} + c^*(k^2\tau\tau')^{-i\mu} \begin{pmatrix} 0 & 0 \\ \sigma_k & 0 \end{pmatrix} + O(k|\tau|, k|\tau'|). \quad (35)$$

The diagonal blocks have only the mixed powers $(\tau/\tau')^{\pm i\mu}$ at zeroth order and hence do not supply the equal-sign soft branch in Eq. (35). The first $k|\tau|$ descendants of the two off-diagonal blocks supply the diagonal entries of Eq. (38); they are required at the first surviving order of the box signal. **There is no SK label on this expression.** For example, the upper-right $+i\mu$ term is $F(-k\tau)G^*(-k\tau')$ in $R^>$ and $G^*(-k\tau)F(-k\tau')$ in $R^<$; both equal $c(k^2\tau\tau')^{i\mu}$. This equality also holds for their analytic momentum descendants. It is compatible with the absence of these powers in the state-independent anticommutator $R^> - R^<$. This property is independent of the choice between the two-Weyl and Majorana descriptions. In particular, the reverse Wightman contraction must use the reversed momentum in its negative-frequency helicity spinors. The two identities

$$h_s(\mathbf{k})h_s^\dagger(\mathbf{k}) = h_{-s}(-\mathbf{k})h_{-s}^\dagger(-\mathbf{k}), \quad (36)$$

$$[u_s(\tau_1)u_s^*(\tau_2)]_{\text{NL}} = -[v_s(\tau_2)v_s^*(\tau_1)]_{\text{NL}} \quad (37)$$

then give $[D_{-+} - D_{+-}]_{\text{NL}} = 0$ in the Majorana description too. The project reference PDF is v1 of Ref. [3], which used a different reverse-momentum expression and introduced a nonzero antisymmetric nonlocal sector. Version v2, Appendix D.3, Eqs. (237)–(240), explicitly includes $-\mathbf{k}$ and establishes the equality of the two nonlocal Wightman contractions. Thus the removal of this particular time-ordering obstruction is a correction to the propagator treatment, not a physical advantage of a Weyl basis. The fermionic minus sign was already present in v1; attributing the discrepancy solely to unsigned Wightman conventions would be incorrect. The old project Majorana box formulas therefore require revision before their final amplitudes can be compared with this calculation. No independent complete recomputation of that old box amplitude is asserted here.

For the first nonzero box signal we also need the first descendants. They are particularly transparent as the following explicit block product:

$$\beta R^{\text{NL}}(\mathbf{k}; \tau, \tau')\Big|_+ = c(k^2\tau\tau')^{i\mu} \begin{pmatrix} \mathbf{1}_2 \\ i\tau\boldsymbol{\sigma} \cdot \mathbf{k} \\ 1+2i\mu \end{pmatrix} \sigma_k \begin{pmatrix} i\tau'\boldsymbol{\sigma} \cdot \mathbf{k} & \\ 1+2i\mu & \mathbf{1}_2 \end{pmatrix} + \dots \quad (38)$$

The other branch is fixed by $R_-^{\text{NL}}(\mathbf{k}; \tau, \tau') = [R_+^{\text{NL}}(\mathbf{k}; \tau', \tau)]^\dagger$. The dots include second-order corrections to the leading entries. They multiply a vanishing zeroth-order Bose-symmetrized hard block and do not affect the first nonzero clock coefficient derived below.

2 Box diagrams and spin contraction

2.1 Diagram counting and four-point integrals

For four labelled external Higgs fields there are $(4 - 1)! = 6$ directed fermion cycles, or three undirected boxes with two orientations each. Each carries $-N_c$, and its symmetry factor is one. Expanding the Dirac determinant gives the same answer: the $1/4$ cancels cyclic rotations of the 4! labelled insertions. There is no extra Majorana factor, and color is counted once.

For $\pi = (i, j, k, l)$ define

$$p_{ij} = q + k_i, \quad p_{jk} = q + k_i + k_j, \quad p_{kl} = q - k_l, \quad p_{li} = q, \quad (39)$$

where this equation is a vector equation. For the four lines of this cycle, abbreviate $R_{W,ij} = R_{W,a_i a_j}(p_{ij}; \tau_i, \tau_j)$, and similarly for the other three pairs. The Feynman rules of Sec. 1.3 were written in the original Ψ_W basis, so their spinor contraction starts as the 4×4 trace

$$\text{tr}_4[\beta_W R_{W,ij} \beta_W R_{W,jk} \beta_W R_{W,kl} \beta_W R_{W,li}]. \quad (40)$$

Insert $R_{W,uv} = U_B R_{uv} U_B^\dagger$ and $\beta_W = U_B \beta U_B^\dagger$ into every factor. Adjacent $U_B^\dagger U_B$ cancel, and cyclicity of the trace removes the remaining pair at the two ends:

$$\begin{aligned} & \text{tr}_4[\beta_W R_{W,ij} \beta_W R_{W,jk} \beta_W R_{W,kl} \beta_W R_{W,li}] \\ &= \text{tr}_4[U_B(\beta R_{ij} \beta R_{jk} \beta R_{kl} \beta R_{li})U_B^\dagger] = \text{tr}_4[\beta R_{ij} \beta R_{jk} \beta R_{kl} \beta R_{li}]. \end{aligned} \quad (41)$$

Both sides remain 4×4 traces. We use the rotated form on the right because the mode functions give R directly. Restoring the branch labels, momenta, time integrals, and external Higgs lines, the exact individual contribution is

$$\mathcal{G}_{ijkl} = -N_c g_t^4 \sum_{a_i, a_j, a_k, a_l = \pm} a_i a_j a_k a_l \int_{-\infty}^0 \prod_{v=i,j,k,l} [\text{d}\tau_v a(\tau_v) G_{a_v}(k_v; \tau_v)] \int \frac{\text{d}^3 q}{(2\pi)^3} \quad (42)$$

$$\times \text{tr}_4[\beta R_{a_i a_j}(p_{ij}; \tau_i, \tau_j) \beta R_{a_j a_k}(p_{jk}; \tau_j, \tau_k) \beta R_{a_k a_l}(p_{kl}; \tau_k, \tau_l) \beta R_{a_l a_i}(q; \tau_l, \tau_i)]. \quad (43)$$

Here $(-i)^4 = 1$ from the four Yukawa vertices.

We can also unpack the same trace into the original two-Weyl components. Let $M_{uv} = \beta_W R_{W,uv}$. Each M_{uv} is a 4×4 matrix made of four 2×2 blocks M_{uv}^{rs} , where $r, s \in \{1, 2\}$ select the upper or lower Weyl block. For two such matrices, ordinary block multiplication and the definition of the trace give the simpler example

$$\text{tr}_4(M_{ij} M_{jk}) = \sum_{r,s=1}^2 \text{tr}_2(M_{ij}^{rs} M_{jk}^{sr}). \quad (44)$$

For the four factors of the box, the same rule gives

$$\begin{aligned}\mathrm{tr}_4(M_{ij}M_{jk}M_{kl}M_{li}) &= \sum_{w_i, w_j, w_k, w_l \in \{1,2\}} \mathcal{S}_{ijkl}^w, \\ \mathcal{S}_{ijkl}^w &= \mathrm{tr}_2[M_{ij}^{w_i w_j} M_{jk}^{w_j w_k} M_{kl}^{w_k w_l} M_{li}^{w_l w_i}].\end{aligned}\quad (45)$$

Each w label selects an upper or lower block index at one vertex; adjacent labels select the block of the connecting M factor. They are neither SK branch labels nor helicities. For example, $w_i w_j w_k w_l = 1112$ selects blocks 11, 11, 12, 21. There are $2^4 = 16$ choices. The tr_2 still sums the two spinor components inside each selected block; summing all 16 results restores the original tr_4 . The table lists blocks of $M = \beta_W R_W$, not blocks of R_W . Because β_W exchanges the two block rows, for example $M_{ij}^{11} = (R_{W,ij})^{21}$. Each term uses exactly the time integrals, SK signs, color factor, and line momenta of Eq. (43). The complete block list is

word	successive M blocks	word	successive M blocks
1111	11,11,11,11	2111	21,11,11,12
1112	11,11,12,21	2112	21,11,12,22
1121	11,12,21,11	2121	21,12,21,12
1122	11,12,22,21	2122	21,12,22,22
1211	12,21,11,11	2211	22,21,11,12
1212	12,21,12,21	2212	22,21,12,22
1221	12,22,21,11	2221	22,22,21,12
1222	12,22,22,21	2222	22,22,22,22

These are 96 algebraic Weyl terms, not 96 distinct box topologies. Keeping the matrix trace performs all these component contractions at once and avoids epsilon-index sign ambiguities.

Figure 1 shows the representative 1234 box. Fixing leg 1, the other five labelled cycles follow by permuting the labels 2, 3, 4 in Eq. (43). This substitution acts on the external momenta, times, and SK labels together; the routing in Eq. (39) then supplies all four internal momenta. Thus the full box contribution is

$$\mathcal{G}_{\mathrm{box}} = \sum_{(j,k,l) \in \mathrm{Perm}(2,3,4)} \mathcal{G}_{1jkl}. \quad (46)$$

The sum has six terms, with no additional permutation factor. For $\mathbf{k}_s = \mathbf{k}_1 + \mathbf{k}_2 \rightarrow 0$, the representative box has two soft lines, $\mathbf{q}$ and $\mathbf{q} + \mathbf{k}_s$, when $\mathbf{q} \sim \mathbf{k}_s$. The relabelled cycles 1234 and 1243 have this same soft region; 1342 and 1432 have two soft lines when $\mathbf{q} + \mathbf{k}_1 \sim \mathbf{k}_s$. The remaining 1324 and 1423 cycles have no two-soft-line cut in the s channel. Consequently the s -channel clock term is the sum of the four listed cut cycles, including their $4 \times 16 = 64$ Weyl-block terms. In the representative box $p_{34} = q - k_4 = q + k_s + k_3$; replacing it by $q + k_3$ before taking the limit would spoil momentum conservation and the first surviving signal. The t - and u -channel results follow by relabelling.

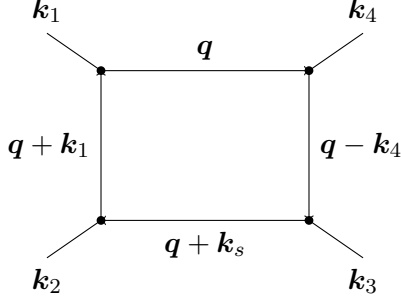


Figure 1: Representative 1234 box, read counterclockwise from the upper-left vertex, with $\mathbf{k}_s = \mathbf{k}_1 + \mathbf{k}_2$. For R_{ij} the charge arrow points from j to i , opposite to the written matrix-cycle order. Relabelling the external legs generates the other five contributions.

2.2 Collapsed limit and Bose-symmetry cancellation

The collapsed kinematics are best organized symmetrically:

$$\mathbf{k}_1 = \mathbf{k}_L + \frac{1}{2}\mathbf{k}_s, \quad \mathbf{k}_2 = -\mathbf{k}_L + \frac{1}{2}\mathbf{k}_s, \quad (47)$$

$$\mathbf{k}_3 = \mathbf{k}_R - \frac{1}{2}\mathbf{k}_s, \quad \mathbf{k}_4 = -\mathbf{k}_R - \frac{1}{2}\mathbf{k}_s, \quad (48)$$

$$\ell = \mathbf{q} + \frac{1}{2}\mathbf{k}_s, \quad \mathbf{K} = \mathbf{q} + \mathbf{k}_s, \quad \mathbf{n}_L = \hat{\mathbf{k}}_L, \quad \mathbf{n}_R = \hat{\mathbf{k}}_R, \quad \mathbf{n}_s = \hat{\mathbf{k}}_s. \quad (49)$$

Write $k_L = |\mathbf{k}_L|$, $k_R = |\mathbf{k}_R|$; at the accuracy of the final answer $k_L = k_{12}/2$, $k_R = k_{34}/2$, where $k_{ij} = k_i + k_j$ is a sum of magnitudes.

For the + clock branch in 1234, the two projected hard matrices are

$$V_L = \begin{pmatrix} \frac{i\tau_1 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} & \mathbf{1}_2 \end{pmatrix} \beta R_{12}(\mathbf{k}_L + \ell; \tau_1, \tau_2) \begin{pmatrix} \mathbf{1}_2 \\ \frac{i\tau_2 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} \end{pmatrix}, \quad (50)$$

$$V_R = \begin{pmatrix} \frac{i\tau_3 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} & \mathbf{1}_2 \end{pmatrix} \beta R_{34}(\mathbf{k}_R + \ell; \tau_3, \tau_4) \begin{pmatrix} \mathbf{1}_2 \\ \frac{i\tau_4 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} \end{pmatrix}. \quad (51)$$

After the Weyl-component sum the entire spin contraction is

$$c^2 (qK)^{2i\mu} \prod_v (-\tau_v)^{i\mu} \text{tr}_2[V_L \sigma_K V_R \sigma_q]. \quad (52)$$

Thus, already before integration,

$$\text{tr}[(\boldsymbol{\sigma} \cdot \mathbf{A}) \sigma_K (\boldsymbol{\sigma} \cdot \mathbf{B}) \sigma_q] = 2[(\mathbf{A} \cdot \hat{K})(\mathbf{B} \cdot \hat{q}) + (\mathbf{A} \cdot \hat{q})(\mathbf{B} \cdot \hat{K}) - (\mathbf{A} \cdot \mathbf{B})(\hat{K} \cdot \hat{q})]. \quad (53)$$

At zeroth order $V_L = -R_{21}(k_L) \propto \boldsymbol{\sigma} \cdot \mathbf{n}_L$ and $V_R \propto \boldsymbol{\sigma} \cdot \mathbf{n}_R$. Their spin trace is odd under either $\mathbf{n}_L \rightarrow -\mathbf{n}_L$ or $\mathbf{n}_R \rightarrow -\mathbf{n}_R$. The sum of the four cut cycles is precisely the independent exchange of the two left and the two right Higgs fields. Therefore

$$\boxed{\mathcal{G}_{s,\text{clock}}^{\text{box}} \Big|_{k_s^{3 \pm 4i\mu}} = 0.} \quad (54)$$

The cancellation follows before any scalar integration: every zeroth-order term contains one factor of each hard direction and is odd under either Higgs-pair exchange. We must therefore keep the first momentum correction at both ends of the cut.

3 Scalar integrals for the hard lines and the soft bubble

3.1 Left and right hard subgraphs

We now calculate the first order of each Bose-symmetrized hard block. Write $x = -k_L\tau_1$, $y = -k_L\tau_2$ and $r = p/k_L$, where $p = |\mathbf{p}|$ denotes the hard-line momentum magnitude in this section. Also write $\sigma_p = \boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$. We need only two SK-summed scalar seeds, corresponding to the 21 and 11 blocks of R :

$$\mathcal{H}_{IJ}(\alpha, \beta; r) = \sum_{a,b=\pm} ab \int_0^\infty dx dy x^{\alpha-1} y^{\beta-1} (1 + iax)(1 + iby) e^{-iax-iby} [R_{IJ}(r; x, y)]_{ab}^{\text{radial}},$$

$$IJ = 21, 11. \quad (55)$$

“Radial” means the coefficient of σ_p in R_{21} or of $\mathbf{1}_2$ in R_{11} ; their mode arguments are rx, ry . Hence these are scalar integrals, with the spin structure already removed. Their finite evaluation is given immediately below in Sec. 3.2.

The useful exact reversal identities are

$$R_{21;ab}(\tau_1, \tau_2) = R_{21;ba}(\tau_2, \tau_1), \quad R_{11;ab}(\tau_1, \tau_2) = -R_{22;ba}(\tau_2, \tau_1). \quad (56)$$

For one left ordering, expanding the endpoint projectors gives

$$V_L = -R_{21} + \frac{i\tau_1}{1 + 2i\mu} \boldsymbol{\sigma} \cdot \mathbf{q} R_{11} - \frac{i\tau_2}{1 + 2i\mu} R_{22} \boldsymbol{\sigma} \cdot \mathbf{K} + O(q^2, k_s^2). \quad (57)$$

All entries here are unapproximated hard SK contractions before expansion of their momentum $\mathbf{p} = \mathbf{k}_L + \boldsymbol{\ell}$. The other ordering has $\mathbf{p} = -\mathbf{k}_L + \boldsymbol{\ell}$. The derivative of the hard line and the descendants of both soft endpoints must both be kept. Eq. (56) shows that the first-order difference between the two external magnitudes gives zero in the sum: the integrated R_{21} block is symmetric in those energies. Also $\int \tau_2 R_{22} = -\int \tau_1 R_{11}$ at equal external magnitudes.

The intermediate momentum expansion makes the two terms in Eq. (64) explicit. Write $h(r) = \mathcal{H}_{21}(i\mu, i\mu; r)$ and $L = \mathbf{n}_L \cdot \boldsymbol{\ell}$. For the ordering $\varepsilon = \pm 1$,

$$|\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}| = k_L + \varepsilon L + O(\ell^2/k_L), \quad (58)$$

$$\boldsymbol{\sigma} \cdot \widehat{\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}} = \varepsilon \boldsymbol{\sigma} \cdot \mathbf{n}_L + \frac{\boldsymbol{\sigma} \cdot (\boldsymbol{\ell} - \mathbf{n}_L L)}{k_L} + O(\ell^2/k_L^2), \quad (59)$$

$$h\left(\frac{|\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}|}{k_L}\right) = h(1) + \frac{\varepsilon L}{k_L} h'(1) + O(\ell^2/k_L^2). \quad (60)$$

The $-R_{21}$ terms of the two orderings therefore add to

$$-\frac{2}{k_L} \boldsymbol{\sigma} \cdot [h(1)\boldsymbol{\ell} + (h'(1) - h(1))\mathbf{n}_L L] + O(\ell^2/k_L^2). \quad (61)$$

The R_{11} and R_{22} descendants in Eq. (57) contribute $-4i\mathcal{H}_{11}(1 + i\mu, i\mu; 1)\boldsymbol{\sigma} \cdot \boldsymbol{\ell}/[k_L(1 + 2i\mu)]$. The common factor $k_L^{-2i\mu}$ comes from $(-\tau_1)^{i\mu}(-\tau_2)^{i\mu}$ after $x = -k_L\tau_1$, $y = -k_L\tau_2$. Finally, the external magnitudes are $|\mathbf{k}_L \pm \mathbf{k}_s/2| = k_L \pm \mathbf{n}_L \cdot \mathbf{k}_s/2 + O(k_s^2/k_L)$. Their first-order shifts cancel under the

two orderings by the time-reversal identity in Eq. (56); equal external energies are imposed only after this sum.

Strip $H^2/(4k_1^3 k_2^3)$ from the left two external lines and vertices. Its integrated projected matrix, summed over its two Higgs orderings, is

$$\mathcal{V}_L = -2k_L^{-1-2i\mu} \boldsymbol{\sigma} \cdot [\mathcal{P}_1(\mu) \boldsymbol{\ell} + \mathcal{P}_2(\mu) \mathbf{n}_L (\mathbf{n}_L \cdot \boldsymbol{\ell})], \quad (62)$$

$$\mathcal{P}_1(\mu) = \mathcal{H}_{21}(i\mu, i\mu; 1) + \frac{2i}{1+2i\mu} \mathcal{H}_{11}(1+i\mu, i\mu; 1), \quad (63)$$

$$\mathcal{P}_2(\mu) = (\partial_r - 1) \mathcal{H}_{21}(i\mu, i\mu; r)|_{r=1}. \quad (64)$$

The two coefficients $\mathcal{P}_1, \mathcal{P}_2$ multiply the independent spin structures $\boldsymbol{\sigma} \cdot \boldsymbol{\ell}$ and $(\boldsymbol{\sigma} \cdot \mathbf{n}_L)(\mathbf{n}_L \cdot \boldsymbol{\ell})$. The $\mathcal{H}_{11}$ term comes from the soft endpoint descendants; the r derivative comes from the hard propagator's momentum dependence. The right result is the same with $(k_L, \mathbf{n}_L) \rightarrow (k_R, \mathbf{n}_R)$ and external prefactor $H^2/(4k_3^3 k_4^3)$. In particular the four-cycle sum is the product of these two already symmetrized blocks; no further factor of two or four is allowed. Formula (64) retains all terms at this order. The absence of the constant term in each block makes the signal a spatial P -wave fermion-pair effect. It adds two powers of k_s to the bubble scaling.

3.2 Evaluation of the scalar hard seeds

Relation to the published Hankel scalar seed. The hard time integrals can also be reduced to the scalar seed of Ref. [1], provided its mass index is analytically continued. This gives a direct route to the literature's single- and double-folded formulae. Set

$$\lambda = \mu - \frac{i}{2}, \quad \rho = i\lambda = \frac{1}{2} + i\mu, \quad x = -k_L \tau_1, \quad y = -k_L \tau_2, \quad r = p/k_L. \quad (65)$$

Define a dimensionless scalar SK kernel $S_{ab}^{(\lambda)}$ from the two Wightman functions

$$S^>(r; x, y) = \frac{\pi}{4} e^{-\pi\lambda} (xy)^{3/2} H_\rho^{(1)}(rx) H_{-\rho}^{(2)}(ry), \quad (66)$$

$$S^<(r; x, y) = \frac{\pi}{4} e^{-\pi\lambda} (xy)^{3/2} H_{-\rho}^{(2)}(rx) H_\rho^{(1)}(ry), \quad (67)$$

with the same SK theta-function ordering as Eq. (26). Using $H_{-\rho}^{(2)} = e^{-i\pi\rho} H_\rho^{(2)}$ and $N_F N_G = \pi/4$ for the mode prefactors in Eq. (15), one finds the *exact* radial identities, for every branch pair,

$$[R_{21}(r; x, y)]_{ab}^{\text{radial}} = \frac{r}{xy} S_{ab}^{(\lambda)}(r; x, y), \quad [R_{11}(r; x, y)]_{ab}^{\text{radial}} = \frac{1}{r} \left(-i\partial_x + \frac{\mu}{x} \right) [R_{21}(r; x, y)]_{ab}^{\text{radial}}. \quad (68)$$

The $1/r$ in the second identity follows from differentiating a mode with argument rx . Differentiating the SK theta functions introduces no delta-function term: the two radial R_{21} Wightman functions agree at $x = y$. Although λ is complex, Eq. (67) is an analytic scalar kernel; no physical function with complex mass is being introduced.

To see precisely which published seeds are needed, let

$$J_{ab}^{p,q}(r) = \int_0^\infty dx dy x^p y^q e^{-iax - iby} S_{ab}^{(\lambda)}(r; x, y). \quad (69)$$

In the units $H = 1$ of Ref. [1], its branch-resolved Hankel seed $\mathcal{J}_{ab}^{p,q}$ obeys the exact normalization map

$$J_{ab}^{p,q}(r) = -ab r^{-5-p-q} \mathcal{J}_{ab}^{p,q}(u, u), \quad u = \frac{2r}{1+r}. \quad (70)$$

The reference includes $-ab$ in the seed definition; neither that sign nor the power of r may be omitted. For $p = \alpha - 2$, $q = \beta - 2$, Eq. (55) becomes

$$\mathcal{H}_{21}(\alpha, \beta; r) = r \sum_{a,b=\pm} ab \left[J_{ab}^{p,q} + ia J_{ab}^{p+1,q} + ib J_{ab}^{p,q+1} - ab J_{ab}^{p+1,q+1} \right]. \quad (71)$$

The four terms correspond to the integer time-power shifts generated by $(1 + ia x)(1 + ib y)$. The SK indices stay explicit until their a, b weights are combined.

The remaining hard coefficient can also be expressed without a new time integral. Put $\mathcal{J}_{ab}^{p,q} = \mathcal{J}_{ab}^{p,q}(1, 1)$, $p_0 = -2 + i\mu + \delta$, and use one common endpoint regulator δ in all seed powers. Equation (71) gives

$$\mathcal{H}_{21}(i\mu + \delta, i\mu + \delta; 1) = - \sum_{a,b=\pm} \sum_{j,l=0}^1 (ia)^j (ib)^l \mathcal{J}_{ab}^{p_0+j, p_0+l}. \quad (72)$$

Integrating the second relation of Eq. (68) by parts yields

$$\begin{aligned} \mathcal{H}_{11}(1 + i\mu + \delta, i\mu + \delta; 1) &= i\delta \mathcal{H}_{21}(i\mu + \delta, i\mu + \delta; 1) \\ &\quad - i \sum_{a,b=\pm} \left[\mathcal{J}_{ab}^{i\mu+\delta, p_0} + ib \mathcal{J}_{ab}^{i\mu+\delta, p_0+1} \right]. \end{aligned} \quad (73)$$

The $i\delta \mathcal{H}_{21}$ term is retained at finite regulator; the complete $\mathcal{H}_{21}$ will be finite, so this term vanishes in the final limit. Individual SK terms do have regulator poles. The integration-by-parts boundary term at $x = 0$ cancels only in the complete SK sum, since for $x < y$ the two branches with the same b have the same Wightman function.

The hard-momentum derivative in $\mathcal{P}_2$ follows from the published scalar bootstrap equation without differentiating a time integral. At $u_1 = u_2 = 1$, the coefficient of its second derivative vanishes. Define

$$T_{ab}^{p,q} = \delta_{ab} e^{-ia\pi(p+q)/2} 2^{-5-p-q} \Gamma(5 + p + q), \quad (74)$$

$$\mathcal{D}_{ab}^{p,q} = \frac{1}{2} \sum_{w=p,q} \frac{\lambda^2 + (w + 5/2)^2}{w + 3} \mathcal{J}_{ab}^{p,q} - \frac{1}{2} \left(\frac{1}{p+3} + \frac{1}{q+3} \right) T_{ab}^{p,q}. \quad (75)$$

Since $du/dr|_{r=1} = 1/2$, $\mathcal{D}_{ab}^{p,q}$ is precisely $d\mathcal{J}_{ab}^{p,q}(u(r), u(r))/dr|_{r=1}$. Combining this with the explicit r^{-5-p-q} in Eq. (70) gives

$$\mathcal{P}_2 = -\text{FP}_{\delta \rightarrow 0} \sum_{a,b=\pm} \sum_{j,l=0}^1 (ia)^j (ib)^l \left[\mathcal{D}_{ab}^{p_0+j, p_0+l} - (5 + 2p_0 + j + l) \mathcal{J}_{ab}^{p_0+j, p_0+l} \right]. \quad (76)$$

Equations (72)–(76), with $\mathcal{P}_1$ from Eq. (64), complete both hard subgraphs in terms of the scalar seed values of Ref. [1]; the right subgraph uses the same two functions.

Regularization and a finite evaluation. There is an essential normalization in Eq. (92) of Ref. [1]: its hypergeometric function is regularized,

$${}_3\tilde{F}_2(a, b, c; d, e; 1) = \frac{{}_3F_2(a, b, c; d, e; 1)}{\Gamma(d)\Gamma(e)}. \quad (77)$$

The earlier numerical implementation omitted these two Gamma factors. Consequently its apparent regulator-path instability was not evidence for an obstruction to the scalar-seed map. A real-index conjugation check alone cannot detect this error. As an independent normalization check, the conformal scalar $\lambda = -i/2$ has $S^>(1; x, y) = xy e^{i(x-y)}/2$. Direct elementary time integration gives $\mathcal{S}_{++}^{0,0} = \mathcal{S}_{--}^{0,0} = -3/16$ and $\mathcal{S}_{+-}^{0,0} = \mathcal{S}_{-+}^{0,0} = 1/32$; the corrected implementation reproduces these values and their unequal-power generalization.

The scalar kernel is invariant under $\lambda \rightarrow -\lambda$, by the Hankel order-reflection identities. In each equal-SK seed choose the sign of λ so that $ai\lambda = 1/2 + i\mu$. Its ordinary ${}_3F_2(1)$ then has convergence balance $1 + i\mu$, whose real part is positive. This is an exact change of representation and needs no additional mass-index regulator.

For this subsection only set $z = i\mu$, and define the single symmetric family

$$\mathcal{Q}_{jl} = \frac{2^{-1-2z-j-l}\Gamma(1+2z+j+l)}{(1+2z+j)(1+2z+l)} {}_3F_2\left(\begin{matrix} 1+2z+j+l, 1+z, 1 \\ 2+2z+j, 2+2z+l \end{matrix}; 1\right). \quad (78)$$

Only $\mathcal{Q}_{00}, \mathcal{Q}_{01}, \mathcal{Q}_{11}, \mathcal{Q}_{02}, \mathcal{Q}_{12}$ are needed. Also abbreviate the finite endpoint contribution by

$$\mathcal{E} = \pi^2 2^{-1-2z} \frac{\Gamma(1+2z)^2}{\Gamma(1+z)^2}. \quad (79)$$

The fully summed hard seeds and momentum derivative are

$$\begin{aligned} \mathcal{H}_{21}(z, z; 1) &= 2 \cosh(\pi\mu)(\mathcal{Q}_{00} + 2\mathcal{Q}_{01} + \mathcal{Q}_{11}) - \mathcal{E}, \\ \mathcal{H}_{11}(1+z, z; 1) &= -2i \cosh(\pi\mu)(\mathcal{Q}_{02} + \mathcal{Q}_{12}), \\ \mathcal{P}_2 &= 2 \cosh(\pi\mu) \sum_{j,l=0}^1 \left[\frac{j(1+2z+j)}{2(1+z+j)} + \frac{l(1+2z+l)}{2(1+z+l)} - (1+2z+j+l) \right] \mathcal{Q}_{jl} \\ &\quad + (1+2z)\mathcal{E} + 2^{-1-2z} \cosh(\pi\mu)\Gamma(1+2z) \frac{4+5z+2z^2}{1+z}. \end{aligned} \quad (80) \quad (81)$$

Together with $\mathcal{P}_1$ in Eq. (64), these are finite, convergent special-function expressions for the complete hard coefficients. No raw mode-power-series truncation or numerical regulator extrapolation is required for the oscillatory signal.

Here is the endpoint cancellation leading to these expressions. For $p_j = -2 + z + \delta + j$, write the Gamma product in the opposite-SK seed as

$$C_{jl}(\delta) = \frac{\Gamma(\delta+j)\Gamma(1+2z+\delta+j)\Gamma(\delta+l)\Gamma(1+2z+\delta+l)}{\Gamma(1+z+\delta+j)\Gamma(1+z+\delta+l)}. \quad (82)$$

After the external-mode factors $(ia)^j(ib)^l$ and the SK branches are summed, its contribution to $\mathcal{H}_{21}$ is $2 \cdot 2^{-1-2z-2\delta-j-l} [\cos(\pi\delta) - 1] C_{jl}(\delta)$. Only $j = l = 0$ has a double pole; therefore

$$\lim_{\delta \rightarrow 0} 2[\cos(\pi\delta) - 1] C_{00}(\delta) = -\pi^2 \frac{\Gamma(1+2z)^2}{\Gamma(1+z)^2}. \quad (83)$$

This is precisely $-\mathcal{E}$ after restoring its power of two. Discarding the poles branch by branch would lose this finite term. The remaining regularized-hypergeometric pieces give $2 \cosh(\pi\mu) \mathcal{Q}_{jl}$. The integration-by-parts insertion in Eq. (73) replaces $j = 0, 1$ by $j = 2$ and contributes $-i$ times the same pieces; it has no double-pole remainder. Finally, substituting p_j in Eq. (75) gives the bracket in Eq. (81). The last term in that equation is the summed bootstrap source, and $(1 + 2z)\mathcal{E}$ is the derivative's endpoint remainder. These steps preserve all factors of r .

The late-time cancellation can also be seen before integration. With $E_a(x) = (1 + iax)e^{-iax}$, the SK sum for $x < y$ is

$$\sum_{a,b} ab E_a(x) E_b(y) R_{ab}(x, y) = [E_+(x) - E_-(x)][E_+(y) R^>(x, y) - E_-(y) R^<(x, y)]. \quad (84)$$

Since $E_+(x) - E_-(x) = -2ix^3/3 + O(x^5)$, the common late-time endpoint is integrable in the hard blocks used here. The early-time BD prescription is still necessary; this identity does not justify integrating undamped oscillations numerically on a truncated interval.

For reference, direct evaluation of Eqs. (80) and (81) gives

μ	$\mathcal{P}_1$	$\mathcal{P}_2$
0.20	$-1.030411081 - 1.109020094i$	$3.163771501 + 0.704629904i$
0.73	$-1.598130708 - 2.346219694i$	$1.923885009 + 1.958596831i$
1.00	$-1.325125325 - 2.665004109i$	$1.137292037 + 2.131408832i$
2.00	$1.901463897 - 2.535666708i$	$-1.119241948 + 1.383844867i$
5.00	$3.921820606 + 1.458261456i$	$-2.047325876 - 0.690483918i$

The corrected branch-resolved calculation converges to these values with a common endpoint regulator. The two representations are checked in `checks/weyl_seed_checks.py`; the physical coefficients are implemented in `checks/weyl_scalar_seed.py`.

3.3 Soft bubble integral

The spin trace of Eq. (53) leaves a polynomial numerator divided by qK . To display it, write the two vectors appearing in the hard blocks as

$$\mathbf{v}_L = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_L (\mathbf{n}_L \cdot \boldsymbol{\ell}), \quad \mathbf{v}_R = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_R (\mathbf{n}_R \cdot \boldsymbol{\ell}). \quad (85)$$

Since $\mathbf{K} = \boldsymbol{\ell} + \mathbf{k}_s/2$ and $\mathbf{q} = \boldsymbol{\ell} - \mathbf{k}_s/2$, the scalar loop integral after the spin contraction is

$$2 \int \frac{d^3 q}{(2\pi)^3} (qK)^{2i\mu-1} \left[2(\mathbf{v}_L \cdot \boldsymbol{\ell})(\mathbf{v}_R \cdot \boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L \cdot \mathbf{k}_s)(\mathbf{v}_R \cdot \mathbf{k}_s) - (\mathbf{v}_L \cdot \mathbf{v}_R)(\ell^2 - k_s^2/4) \right]. \quad (86)$$

Its mass dimension is $5 + 4i\mu$. We now evaluate it by scalar convolutions, following the same procedure as for the vector box. The only required scalar formula is

$$F_0(A, B) = \frac{\Gamma(A + B - \frac{3}{2})\Gamma(\frac{3}{2} - A)\Gamma(\frac{3}{2} - B)}{\Gamma(A)\Gamma(B)\Gamma(3 - A - B)}, \quad (87)$$

$$\int \frac{d^3 q}{(2\pi)^3} (q^2)^{-A} (K^2)^{-B} = \frac{k_s^{3-2A-2B}}{(4\pi)^{3/2}} F_0(A, B). \quad (88)$$

The right-hand side is the analytic continuation of the nonanalytic part; UV-dependent polynomial contact terms are excluded.

For the reduction only, measure q, K, ℓ in units of k_s , put $u = q^2$, $v = K^2$, and set $\nu = 1/2 - i\mu$. Then

$$\ell^2 = (u + v)/2 - 1/4, \quad \ell_{\parallel} \equiv \boldsymbol{\ell} \cdot \mathbf{n}_s = (v - u)/2, \quad \ell_{\perp}^2 = \ell^2 - \ell_{\parallel}^2. \quad (89)$$

Azimuthal averaging replaces two transverse components by $\ell_{\perp}^2 \delta_{ij}^{\perp}/2$ and four by $\ell_{\perp}^4 (\delta_{ij}^{\perp} \delta_{kl}^{\perp} + \delta_{ik}^{\perp} \delta_{jl}^{\perp} + \delta_{il}^{\perp} \delta_{jk}^{\perp})/8$, where $\delta_{ij}^{\perp} = \delta_{ij} - n_{si} n_{sj}$. The numerator in Eq. (86) consequently becomes a polynomial in u, v and the three external angles. Each monomial is integrated by

$$u^m v^n \mapsto \frac{F_0(\nu - m, \nu - n)}{(4\pi)^{3/2}}, \quad (90)$$

$$\frac{F_0(\nu - m, \nu - n)}{F_0(\nu, \nu)} = \frac{(3/2 - \nu)_m (3/2 - \nu)_n (1 - \nu)_m (1 - \nu)_n}{(5/2 - 2\nu)_{m+n} (3 - 2\nu)_{m+n}}. \quad (91)$$

Only nonnegative integers m, n with $m + n \leq 4$ occur. Thus the bubble is a finite Gamma sum, and all its terms reduce to the same $F_0(\nu, \nu)$ times rational functions. For example the coefficient of $\mathcal{P}_1^2$ in the square bracket of Eq. (86) is $\ell^4 + \ell^2/4 - \ell_{\parallel}^2/2$, whose integral is

$$\frac{F_0(\nu, \nu)}{(4\pi)^{3/2}} \frac{3(\nu - 1)(2\nu - 3)}{8(\nu - 2)(4\nu - 7)(4\nu - 5)}. \quad (92)$$

The $\mathcal{P}_1 \mathcal{P}_2$ and $\mathcal{P}_2^2$ terms are evaluated by the same two steps. Their fully reduced coefficients are displayed directly in Eq. (100), without introducing separate symbols for each tensor moment. The azimuthal average introduces only the five angle monomials $1, c_1^2 + c_3^2, c_1^2 c_3^2, \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3$, and $\sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3)$, where $c_i = \cos \theta_i$. Applying Eq. (91) to their coefficients reproduces Eq. (100).

4 Nonlocal signal

4.1 Oscillatory signal and angular coefficients

Choose the angle convention

$$\mathbf{n}_L = (\sin \theta_1, 0, \cos \theta_1), \quad \mathbf{n}_R = (\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3), \quad \mathbf{n}_s = (0, 0, 1). \quad (93)$$

The leading nonzero oscillatory term, summed over the four s -cut boxes, is

$$\begin{aligned} \langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle' |_{s, \text{clock}} = & - \frac{N_c g_t^4 H^4 k_s^5}{16 k_1^3 k_2^3 k_3^3 k_4^3 k_L k_R} \\ & \times \left[\left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2i\mu} \mathcal{C}(\mu; \theta_1, \theta_3, \phi_3) + \text{c.c.} \right]. \end{aligned} \quad (94)$$

The coefficient is given directly in the angular basis of the vector-box note:

$$\mathcal{C} = A_1 + A_2(\cos^2 \theta_1 + \cos^2 \theta_3) + A_3 \cos^2 \theta_1 \cos^2 \theta_3$$

$$+ B_1 \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3 + D_1 \sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3). \quad (95)$$

All five coefficients share one soft-bubble normalization, including the two soft-mode coefficients c^2 , the factor 4 from the two Bose-summed hard blocks, the factor 2 from the Pauli trace, and the phase from the chosen momentum ratio:

$$\mathcal{B}(\mu) = \frac{2^{1+4i\mu} \Gamma(\frac{1}{2} - i\mu)^4 F_0(\nu, \nu)}{32\pi^2 (4\pi)^{3/2} (\nu - 3)(\nu - 2)(4\nu - 7)(4\nu - 5)}, \quad \nu = \frac{1}{2} - i\mu. \quad (96)$$

In particular, the natural hard scaling is $(k_s^2/(k_L k_R))^{2i\mu}$. Since $4k_{12}k_{34} = 16k_L k_R$ at this order, using the vector note's ratio transfers $16^{2i\mu}$ into the coefficient. Thus $8c^2 16^{2i\mu} = 2^{1+4i\mu} \Gamma(1/2 - i\mu)^4/\pi^2$, which fixes both the phase and the overall factor in Eq. (96). Combining the bubble with the two hard coefficients of Eq. (64) gives the final mass dependence:

$$A_1 = \mathcal{B}(\nu - 1)(2\nu - 3) \left[4(\nu - 3)(3\mathcal{P}_1^2 + 2\mathcal{P}_1\mathcal{P}_2) + (2\nu - 5)\mathcal{P}_2^2 \right], \quad (97)$$

$$A_2 = -\mathcal{B}\mathcal{P}_2^2(\nu - 1)(2\nu - 3)^2, \quad A_3 = -3A_2, \quad (98)$$

$$B_1 = -2\mathcal{B}\mathcal{P}_2^2(\nu - 1)(\nu - 2)(2\nu - 3), \quad (99)$$

$$D_1 = \mathcal{B}\mathcal{P}_2^2(\nu - 2)(2\nu - 5)(2\nu - 3). \quad (100)$$

Thus the answer requires only two hard functions $\mathcal{P}_1, \mathcal{P}_2$ and one explicit bubble factor $\mathcal{B}$. The coefficients are complex and include the clock phase. The full answer is even under either Higgs-pair exchange and under parity. Its engineering dimension is -5 , as required for the primed four-point function. The powers of H follow from four rescaled Weyl Yukawa vertices and four external Higgs contractions; no extra factor of H or a second color/orientation multiplicity is implicit. Higher soft powers and analytic contact terms are not displayed.

4.2 Nonoscillatory contribution

Taking opposite soft branches gives a nonoscillatory k_s^3 term. The hard blocks are R_{11} and $-R_{22}$; Eq. (56) makes their integrated values equal to the real seed $\mathcal{H}_{11}(-i\mu, +i\mu; 1)$. The leading soft matrices are now opposite triangular blocks, so they project the diagonal hard blocks. Each hard end therefore has a Bose-even zeroth-order term. There are two soft assignments and two Higgs orderings at each end, giving $2 \times 2 \times 2$ contributions; the spin trace adds $2\hat{q} \cdot \hat{K}$. The soft coefficients give $|c|^2 = 1/(4 \cosh^2 \pi\mu)$. These factors fix the normalization of the mixed term independently of the oscillatory branch. The scalar bubble in this case is

$$\int \frac{d^3 q}{(2\pi)^3} (q^2)^{-i\mu} (K^2)^{i\mu} \hat{q} \cdot \hat{K} = \frac{k_s^3}{(4\pi)^{3/2}} \frac{2\sqrt{\pi}}{3} \mu(1 + \mu^2) \coth(\pi\mu). \quad (101)$$

After both soft branches and all four s -cut boxes are summed, the result is

$$\langle \delta h^4 \rangle' \Big|_{s, \text{mixed}} = \frac{N_c g_t^4 H^4 k_s^3}{16 \prod_i k_i^3} \frac{2\mu(1 + \mu^2)}{3\pi \sinh(2\pi\mu)} \left[\mathcal{H}_{11}(-i\mu, +i\mu; 1) \right]^2. \quad (102)$$

It is independent of angles at this order. Equation (102) is the leading nonoscillatory nonanalytic term, whereas Eq. (94) is the leading mass-dependent clock signal. At $\mu = 0$ the separated

branches coalesce; the limit must be taken after combining the relevant terms. Here $k_s^3 = (\mathbf{k}_s^2)^{3/2}$ is nonanalytic as a function of the soft momentum components, even though it has no mass-dependent oscillation. It cannot be discarded as a polynomial contact term.

For evaluation of the remaining mixed seed, the same radial identity and integration by parts give the general formula

$$\begin{aligned} \mathcal{H}_{11}(\alpha, \beta; 1) &= [\mathrm{i}(\alpha - 1) + \mu] \mathcal{H}_{21}(\alpha - 1, \beta; 1) \\ &\quad - \mathrm{i} \sum_{a,b=\pm} \left[\mathcal{S}_{ab}^{\alpha-1, \beta-2} + \mathrm{i}b \mathcal{S}_{ab}^{\alpha-1, \beta-1} \right]. \end{aligned} \quad (103)$$

Set $\alpha = -\mathrm{i}\mu + \delta$, $\beta = \mathrm{i}\mu + \delta$, perform the complete sum, and then take $\delta \rightarrow 0$ using the corrected regularized seeds. The conjugation relation $\mathcal{H}_{11}(\alpha, \beta; 1)^* = \mathcal{H}_{11}(\beta^*, \alpha^*; 1)$ proves the mixed seed is real. The code evaluates this limit by fourth-order extrapolation, with step-size and precision checks; the clock coefficients above instead use the analytic limit. For example, $\mathcal{H}_{11}(-0.73\mathrm{i}, +0.73\mathrm{i}; 1) = -1.388500280198814$; halving the extrapolation step changes this value by less than 2×10^{-15} .

Checks and scope. The accompanying scripts check the mode equations, normalization, signed SK contractions, soft descendants, spin traces, scalar Gamma reductions, and hard momentum expansion. The hard-seed checks include independent conformal-scalar time integrals, the exact finite SK sum, regulator convergence, and precision stability. The former hard-seed normalization error is corrected by Eq. (77); the scaling, diagram multiplicities and five angular coefficients retain their displayed form. These checks concern the leading collapsed nonanalytic terms of the constant-mass, zero-chemical-potential Yukawa box in de Sitter with a light external Higgs. They are not a numerical integration of the full unexpanded box, a computation of its renormalized analytic contact terms, or a Higgs-to-curvature conversion. The gauge-boson note supplies the factorization workflow and angular basis. Its zeroth-order hard-line replacement is insufficient for this fermion clock term: the two first descendants and the hard momentum derivative are essential. The loop factorization follows Ref. [4].

References

- [1] Z. Qin and Z.-Z. Xianyu, “Closed-Form Formulae for Inflation Correlators,” JHEP **07** (2023) 001, [arXiv:2301.07047](#), especially the branch-resolved Hankel seed and its folded limits.
- [2] I. I. Cotăescu, “Integral representation of the Feynman propagators of the Dirac fermions on the de Sitter expanding universe,” Eur. Phys. J. C **78**, 769 (2018), [doi:10.1140/epjc/s10052-018-6258-2](#).
- [3] J. You, L. Song, C. Han, H.-J. He, X. Chen and Z.-Z. Xianyu, “Cosmological Collider Signatures from Right-Handed Neutrino Loop,” [arXiv:2605.21419v2](#), Appendix D.3.
- [4] Z. Qin and Z.-Z. Xianyu, “Inflation Correlators at the One-Loop Order: Nonanalyticity, Factorization, Cutting Rule, and OPE,” JHEP **09** (2023) 116, [arXiv:2304.13295](#).

- [5] S. Lu, Y. Wang and Z.-Z. Xianyu, “A Cosmological Higgs Collider,” JHEP **02** (2020) 011, [arXiv:1907.07390](#).